

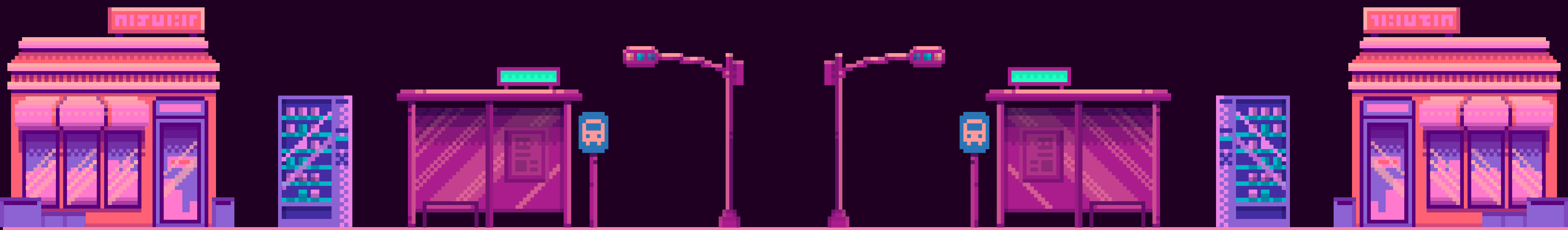
CHOOSE YOUR
FIGHTER:



SPRING-AI VS

LANGCHAIWU

MALAVIKA BALAMURALI

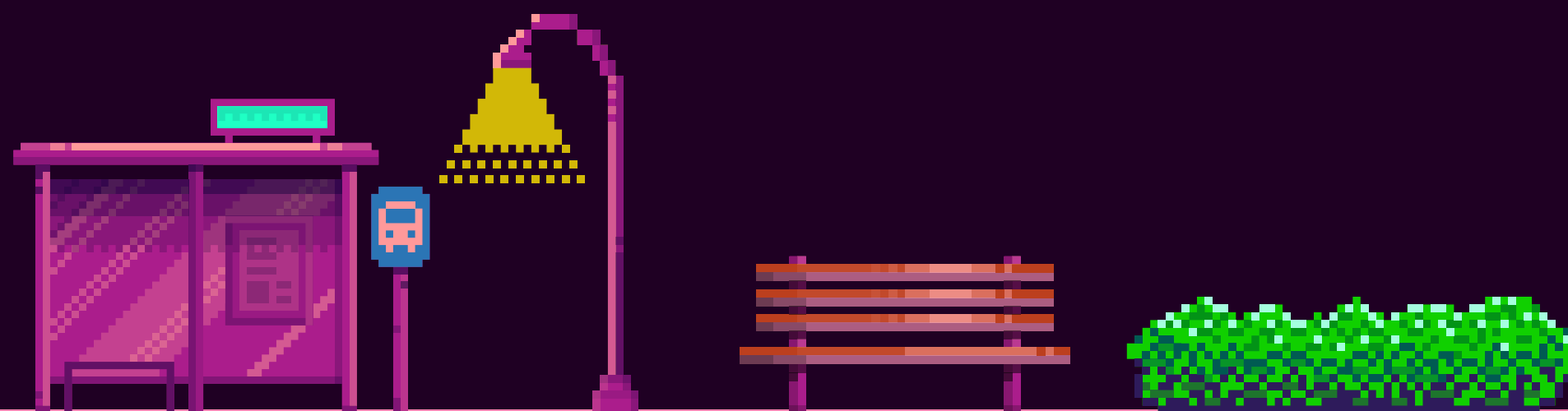


THE SITUATION

- Our world relies on AI more and more.
- As a result, Python has grown in popularity.
- Java programmers feel left behind and left out.



spring[®] AI



INTRODUCTION

Malavika Balamurali

- CS major @ KSU
- Junior
- With Java since version 8



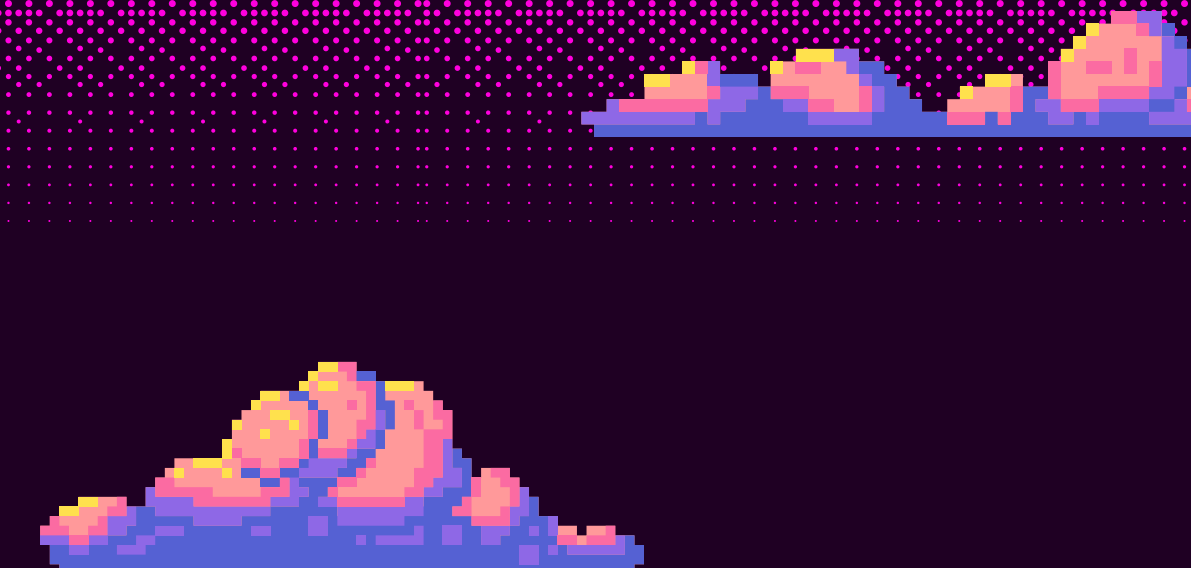
Mbala658



bvika0207@outlook.com



WHAT IS SPRING AI?



01

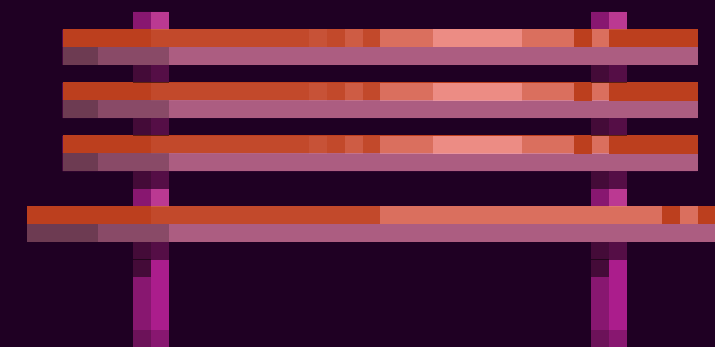
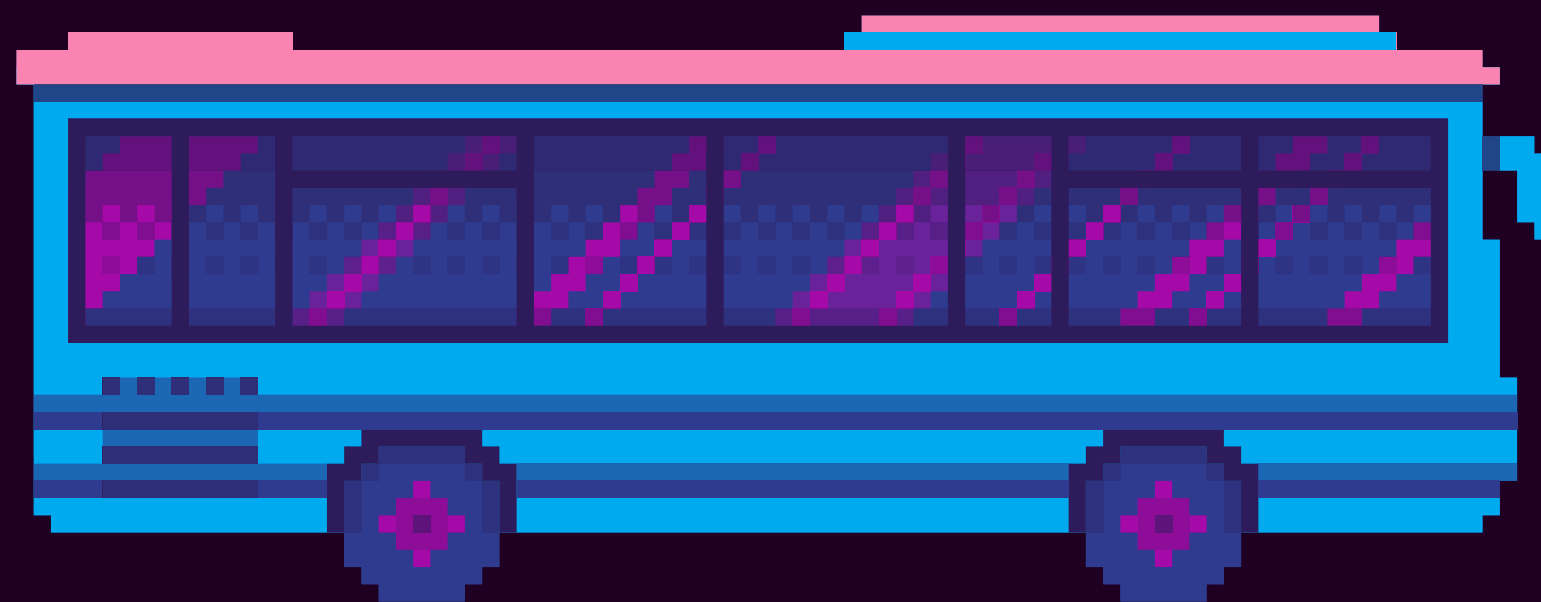
Project within the
Spring Framework
ecosystem

02

Provides tools to help
developers integrate AI
capabilities, especially LLMs
and Generative AI services, into
Java applications.

03

Makes it easier for Java
developers to use tools like
OpenAI, Hugging Face, Azure
OpenAI and local models.



WHAT IS LANGCHAIN4J?

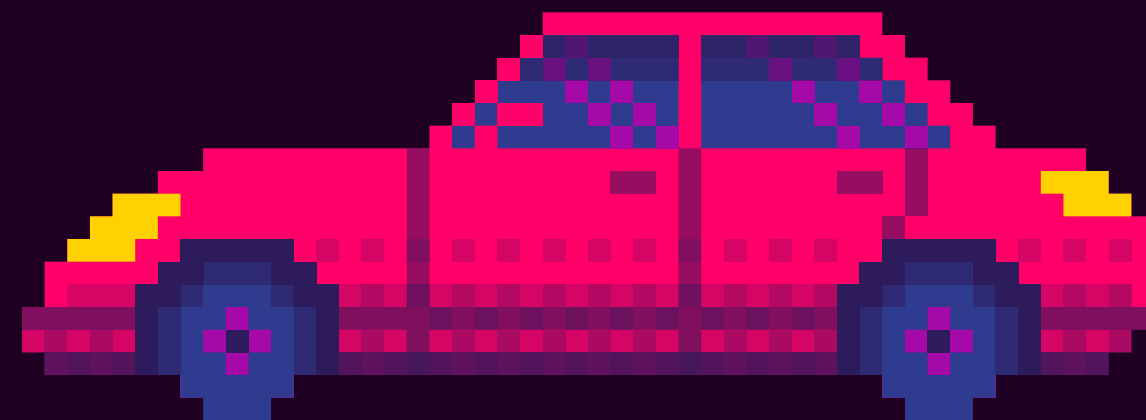
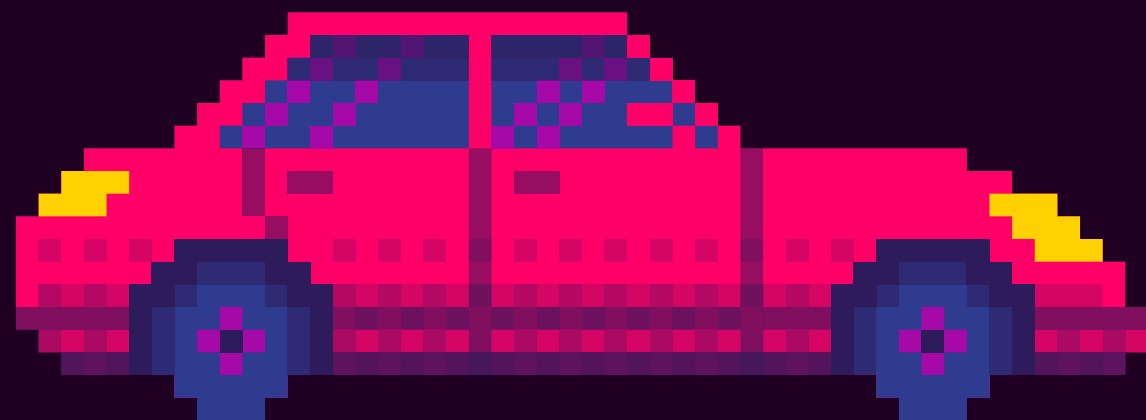
LangChain4j is a Java library inspired by Python's LangChain.



Provides abstractions and tools (prompt templates, memory, vector stores, function calling/tools, etc.) for building AI applications in Java.



It is more model/provider agnostic and framework-agnostic (though it has integrations).

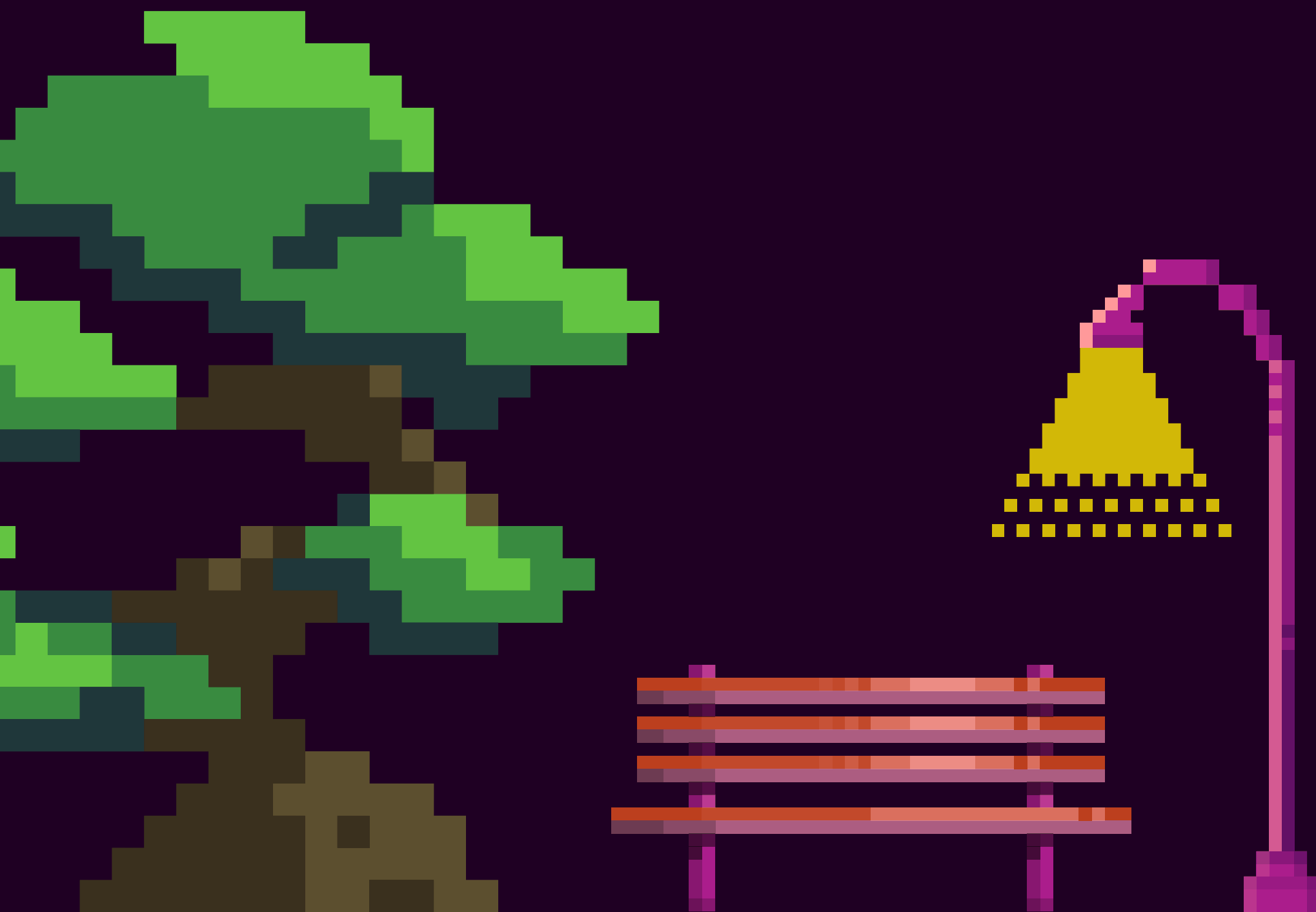


HOW DOES

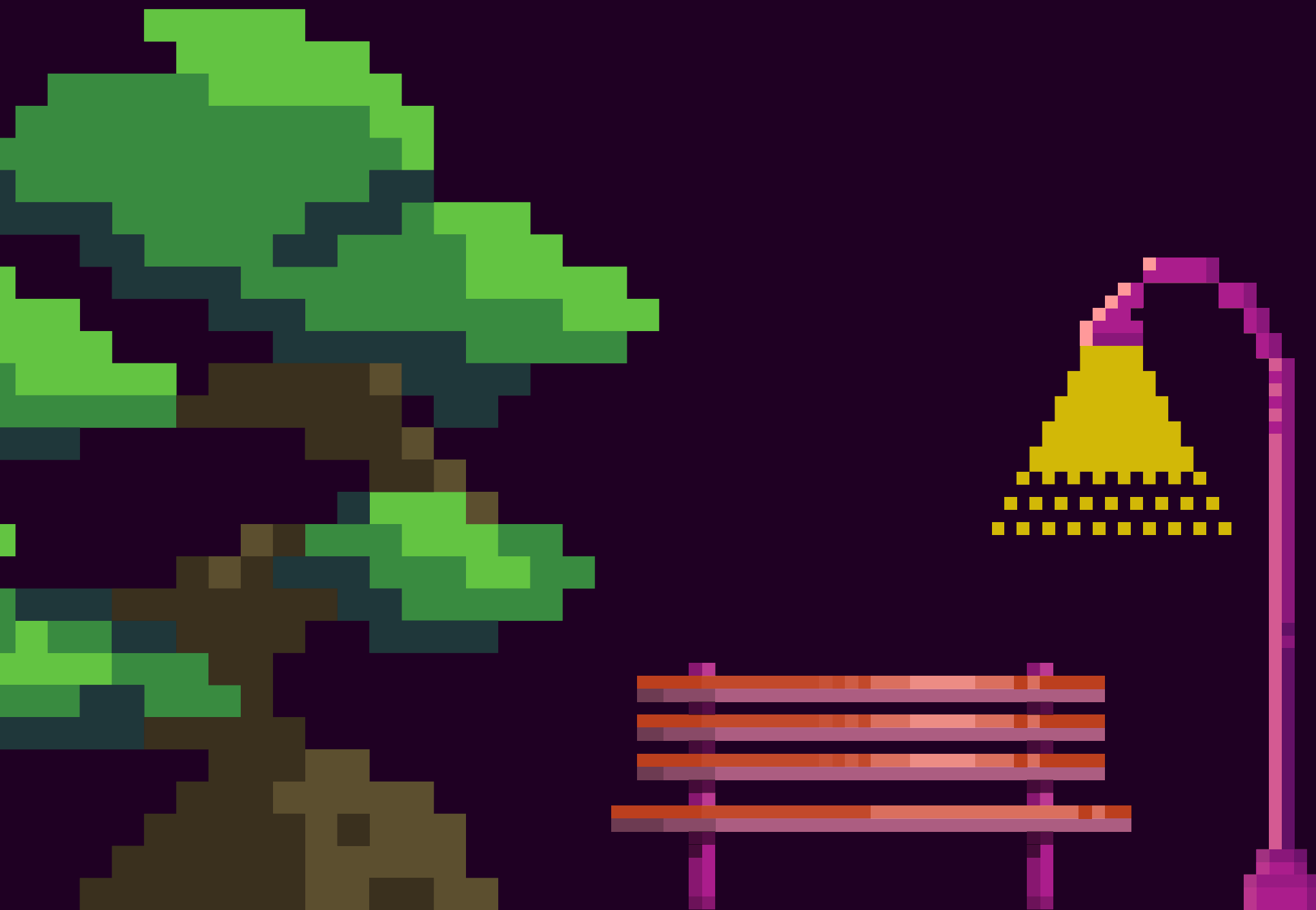
IT WORK

SpringAI

1. Spring Boot Starts up
2. Auto-configuration registers both ChatModel and EmbeddingModel
3. Configuration is loaded from application.yml/application.properties and environment variables
4. Application injects model bean
5. Runtime:
 - a. Prompt template resolved
 - b. Parameters bound
 - c. Memory injected
6. HTTP request constructed internally
7. Response mapped to Spring AI abstractions



HOW DOES IT WORK

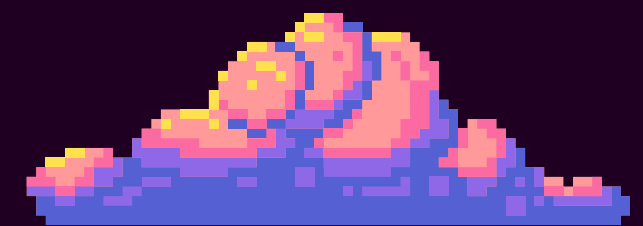
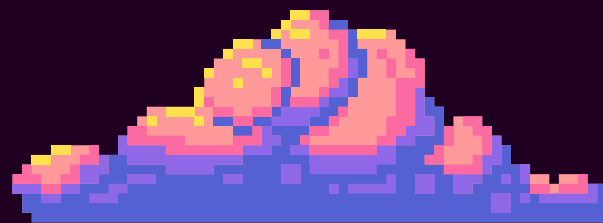


Langchain4j

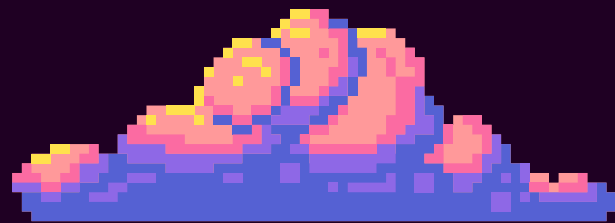
1. Application starts
2. Developer instantiates:
 - a. Model
 - b. Memory
 - c. Tools
3. Components wired explicitly in code
4. Prompt constructed programmatically
5. Memory explicitly applied
6. HTTP request built in user-visible code path
7. Response mapped directly to domain types

HOW ARE DEPENDENCIES

HANDLED?



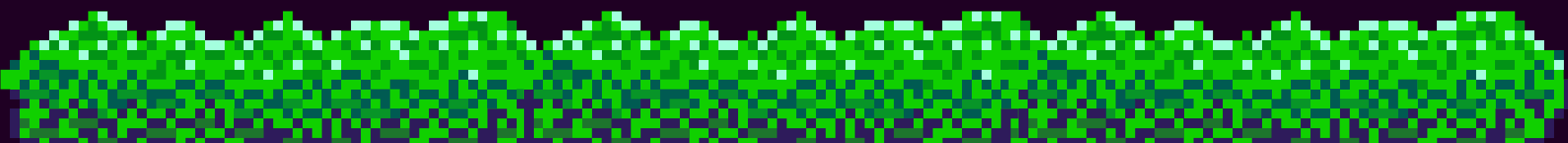
Spring AI



- LLMs as injectable beans
- Auto-config via starters
- Config externalized
- Late binding via profiles

LangChain4J

- Manual wiring of components
- Constructor-level dependency control
- Compile-time visibility of dependencies
- No runtime magic



WHY DOES IT MATTER?

- AI ≠ just another dependency
- Impacts architecture, testing, and ops
- Different frameworks → different trade-offs
- Choice affects where complexity lives

A SPLIT OF RESPONSIBILITIES

Concern	Spring AI	LangChain4j
Model lifecycle	Framework	Application
Prompt loading	Framework	Application
Memory wiring	Framework	Application
Provider switching	Config	Code

PROVIDER ABSTRACTION

- OpenAI, Azure, Bedrock, local models
- Vendor-specific APIs differ
- Abstraction reduces coupling

PROMPT LIFE CYCLE

Problems

- Prompts change often
- Need parameterization
- Need versioning
- Need environment tuning

Approaches

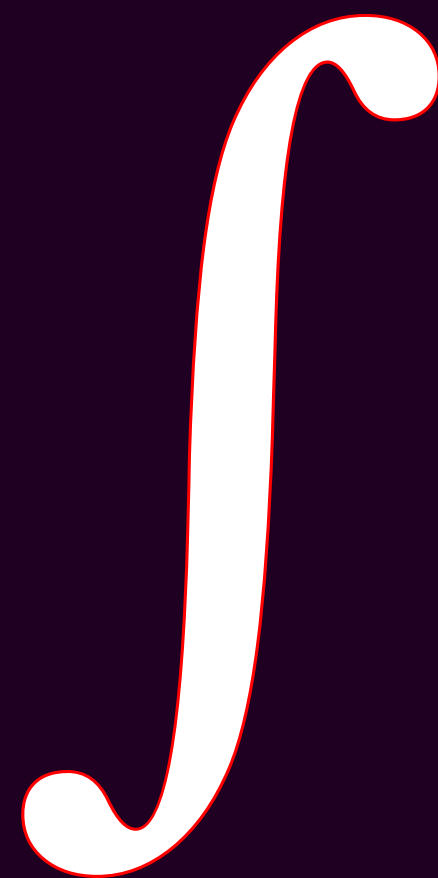
- Spring AI: prompts as resources
- LangChain4j: prompts as code

WHA...
A...

LET'S TAKE A BREATHER

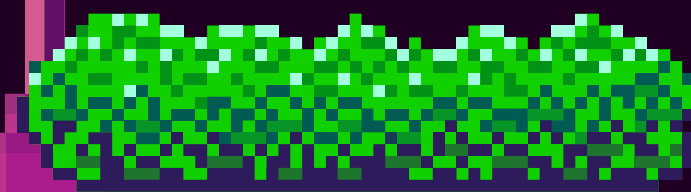
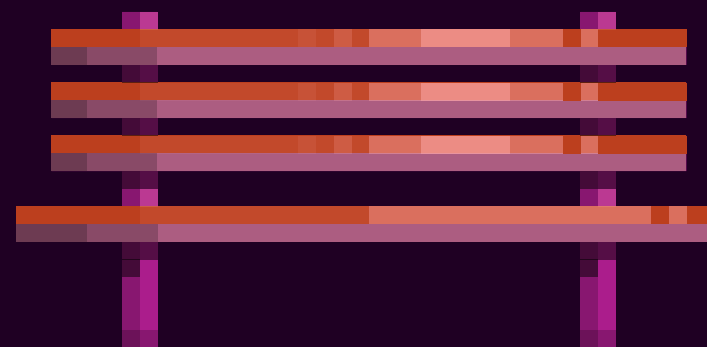
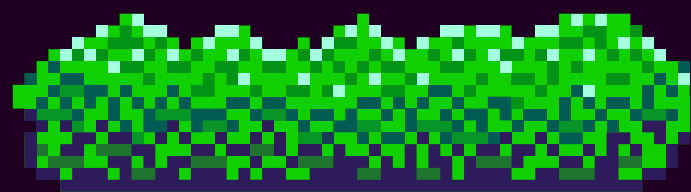
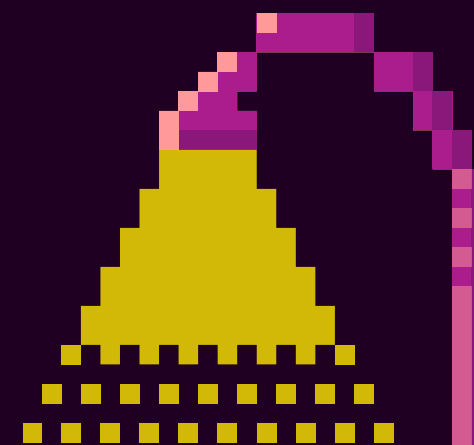
DEMO TIME

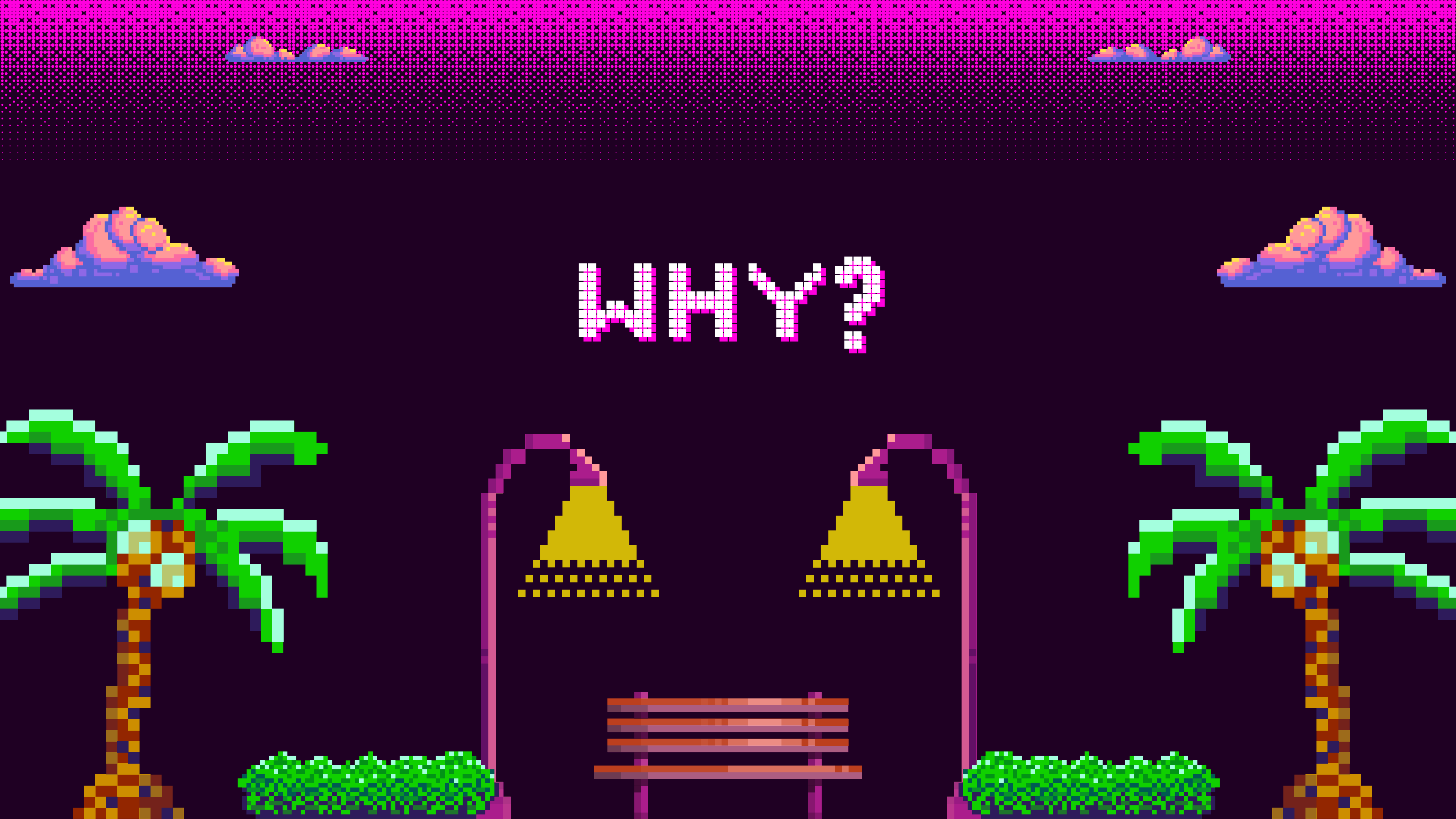




CATEGORIES:

- Modularity
- Architecture style
- Integration
- Scalability
- Reusability
- Configuration
- Documentation





Dependencies

	SpringAI	LangChain4j
Ollama		<pre><dependency> <groupId>dev.langchain4j</groupId> <artifactId>langchain4j-ollama</artifactId> <version>1.10.0</version> </dependency></pre>
Claude		<pre><dependency> <groupId>dev.langchain4j</groupId> <artifactId>langchain4j-anthropic</artifactId> <version>1.10.0</version> </dependency></pre>
OpenAI		<pre><dependency> <groupId>dev.langchain4j</groupId> <artifactId>langchain4j-openai</artifactId> </dependency></pre>

RESOURCES

SpringAI

LangChain4j

SO, WHICH ONE SHOULD
YOU USE?



IT

DEPENDS

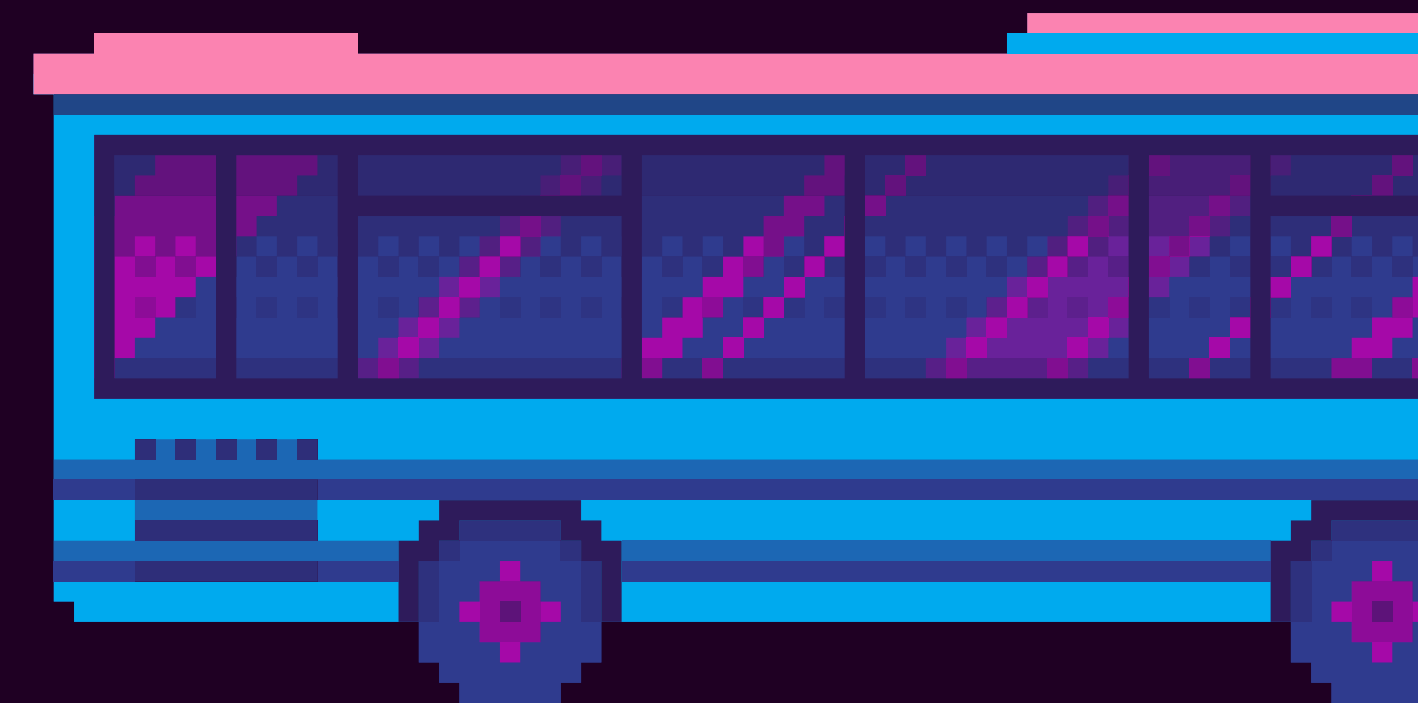
01

Langchain4j and SpringAI have their own pros and cons. It's up to the team programming the application to decide which one suits them better.

02

If you're more into Spring, you can use SpringAI. If you like using Quarkus or just vanilla Java, Langchain4j may work better.

Remember, it's all about the application and what it will do and what it can work with.





THANK YOU